

Q4(a)

Two waves of the same type travelling in opposite directions superpose on each other. [1]

They have the same speed, amplitude and frequency. [1]

Q4(b)(i) $f = v/\lambda$ [1]

$= 330/0.18 = \mathbf{1830 \text{ Hz}}$ [1]

Q4(b)(ii)

$T = 1/f = 5.46 \times 10^{-4} \text{ s}$ [1]

1 period is 27 squares $= 27/5 \text{ cm} = 5.4 \text{ cm}$.

Time base setting $= 5.46 \times 10^{-4} / 5.4 = \mathbf{1.01 \times 10^{-4} \text{ s cm}^{-1}}$ [1]

Q4(b)(iii)

$$I = kA^2$$

$I/2 = k(A')^2$, the new amplitude A' is $A/\sqrt{2}$.

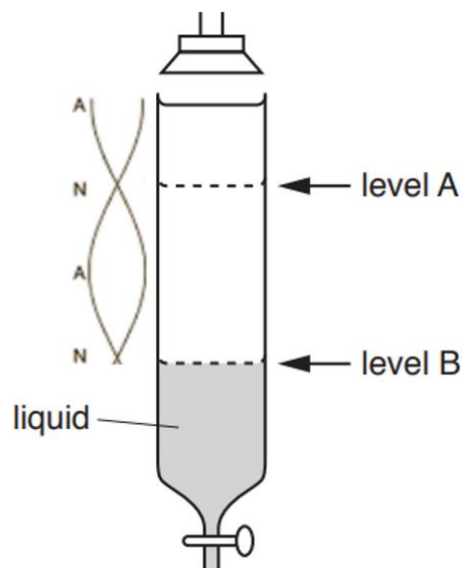
12 small squares correspond to A . Therefore, $A/\sqrt{2}$ corresponds to 8.5 small squares.

Draw sine shape with same period, [1]

amplitude 8.5 small squares. [1]

Q4(c)(i)

Distance AB is half a wavelength $= 0.18/2 = \mathbf{0.090 \text{ m}}$ [1]



(ii) N at position A and B. [1]

